Excel Formulas & Functions

Dataset Used: Order_Revenue_Dataset

1. Statistical Functions:

Functions	Description	Formula
Mean (Average)	*Sum of All Values / Total Number of Values*	=AVERAGE(number1,[number2],…)
Median	*Middle value in the sorted dataset*	=MEDIAN(number1, [number2],…)
Mode (Single)	*Most frequently occurring value*	=MODE.SNGL(number1,[number2],…)
Mode (Multiple)	*All modes in a dataset (if more than one mode)*	=MODE.MULT(number1,[number2],…)

Formula & Output:

Sample Formula	Data Output [Unit Price]
"=AVERAGE(Revenue_DB[Unit Price ($)])"	**206.99**
"=MEDIAN(Revenue_DB[Unit Price ($)])"	**49.99**
"=MODE.SNGL(Revenue_DB[Unit Price ($)])"	**29.99**
"=MODE.MULT(Revenue_DB[Unit Price ($)])"	**29.99**
	29.99

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2. Mathematical & Aggregate Functions:

Functions	Description	Formula
SUM	***Sum numbers***	SUM(number1,[number2],…)
SUMIF	***Sum with condition***	SUMIF(range,criteria,[sum_range])
MIN	***Finds Minimum***	MIN(number1,[number2],…)
MAX	***Finds Maximum***	MAX(number1,[number2],…)
COUNT	***Counts numbers***	COUNT(value1,[value2],…)
COUNTBLANK	***Counts empty cells***	COUNTBLANK(range)
COUNTIF	***Count with condition***	COUNTIF(range,criteria)
AVERAGEIF	***Average with condition***	AVERAGEIF(range,criteria, [average_range])
AVERAGEIFS	***Average with multiple condition***	AVERAGEIFS(average_range, criteria_range1,criteria1, [criteria_range2, criteria2],…)

Formula & Output:

- **SUM Function:**
  - ***"=SUM(Revenue_DB[Total Revenue ($)])"***
- **SUMIF Function:**
  - ***"=SUMIF(Revenue_DB[Region],"East",Revenue_DB[Total Revenue ($)])"***
- **MIN Function:**
  - ***"=MIN(Revenue_DB[Total Revenue ($)])"***
- **MAX Function:**
  - ***"=MAX(Revenue_DB[Total Revenue ($)])"***
- **COUNT Function:**
  - ***"=COUNT(Revenue_DB[Total Revenue ($)])"***
- **COUNTBLANK Function:**
  - ***"=COUNTBLANK(Revenue_DB[Order ID])"***

- **COUNTIF Function:**
  - ***"=COUNTIF(Revenue_DB[Region],"West")"***

- **AVERAGEIF Function:**
  - ***"=AVERAGEIF(Revenue_DB[Region],"North", Revenue_DB[Total Revenue ($)])"***

- **AVERAGEIFS Function:**
  - ***"=AVERAGEIFS(Revenue_DB[Total Revenue ($)],Revenue_DB[Region],"South",Revenue_DB[Category],"Furniture")***

Functions	Data output [Total Revenue], [Region], [Category] & [Order ID]
SUM	248886.22
SUMIF	75950.21
MIN	39.96
MAX	51934.6
COUNT	50
COUNTBLANK	5
COUNTIF	12
AVERAGEIF	2825.504545
AVERAGEIFS	6772.2625

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3. Logical Functions:

Functions	Description	Formula
IF	***Conditional test***	IF(logical_test,[value_if_true],[value_if_false])
AND	***Returns TRUE if all conditions are TRUE***	AND(logical1,[logical2],…)
OR	***Returns TRUE if any condition is TRUE***	OR(logical1,[logical2],…)
NOT	***Reverses a logical value (TRUE becomes FALSE)***	NOT(logical)

Sample Formulas:

Functions	Sample Formula
IF	***"=IF([@[Units Sold]]>30,"High Sales","Low Sales")"***
AND	***"=AND([@[Units Sold]]>30,[@[Total Revenue ($)]]>=1000)"***
IF(AND)	***"=IF([@AND],"Best Sales","Worst Sales")"***
OR	***"=OR([@[Units Sold]]>30,[@[Total Revenue ($)]]>=1000)"***
NOT	***"=NOT([@[Discount (%)]]>10)"***

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4. Text Functions:

Functions	Description	Formula
RIGHT	***Right characters***	RIGHT(text,[num_chars])
LEFT	***Left characters***	LEFT(text,[num_chars])
MID	***Middle characters***	MID(text,start_num,num_chars)
UPPER	***Convert to UPPERCASE***	UPPER(text)
LOWER	***Convert to lowercase***	LOWER(text)
PROPER	***Capitalize first letters***	PROPER(text)
LENGTH	***Length of text***	LEN(text)

Sample Formulas:

Functions	Sample Formula
RIGHT	***"=RIGHT([@[Order ID]],4)"***
LEFT	***"=LEFT([@[Order ID]],3)"***
MID	***"=MID([@[Order ID]],1,4)"***
UPPER	***"=UPPER([@Category])"***
LOWER	***"=LOWER([@Product])"***
PROPER	***"=PROPER([@LOWER])"***
LENGTH	***"=LEN([@UPPER])"***
